

A. List the ways to assess a patient for the presence of residual neuromuscular blockade. (30%)

B. Discuss methods to reduce the risk of residual neuromuscular blockade (70%)

(A)

Clinically:

History

- Dyspnea, muscle weakness
- Review anaesthesia chart: last dose of NMBA, reversal dose (whether adequate for weight)

Examination

- Poor respiratory effort: rapid shallow breathing, poor air entry, minimal chest movement, use of accessory muscles of respiration

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- Ptosis
- Twitchy appearance

Unable to sustain head lift ≥ 5 seconds, weak hand grip Inability to swallow secretions

- Investigations: desaturation, low pO_2 /raised pCO_2 on ABG

Depth of neuromuscular blockade monitoring:

- Quantitative (train of four ratio < 0.9): acceleromyography, electromyography, mechanomyography
- Qualitative: conventional peripheral nerve stimulator -- TOF count < 4 with fade, double burst stimulation

with fade, post tetanic count low with fade

(B)

Avoid use of NMBA

- Choose regional anaesthesia where possible

§ Dependent on nature of surgery, may not be possible if patient refuses, prolonged surgery

■

Identify patients at high risk of residual NMB

■

electrolyte abnormalities

- Reduced clearance: hepatic/renal dysfunction, hypothermia, elderly

Dosing of NMB

- Alter doses in patients with altered physiology and metabolism as listed above

§ No clear guidelines regarding dose adjustments

§ Less problematic with atracurium due to Hoffman's degradation (organ independent metabolism)

- Avoid infusions, or clear communication with surgeons regarding remaining op time to facilitate early cessation of infusion

Use depth of NMB monitoring

- Obtain baseline results post induction pre NMBA administration, removes doubt whether PNS working functioning adequately at end of op
- Guides NMBA top ups, avoid excessive dosing and overly deep blockade (aim TOF 1-2)

■ Quantitative monitoring preferable to qualitative

§ More accurate but requires specialised equipment

Avoid agents that potentiate NMB

■ Volatiles, aminoglycosides, MgSO₄

§ Ensure adequate removal of volatile agents prior to extubation/removal of LMA

- Full reversal of NMBA at end of op to ensure TOFR >0.9

■ Weight adjusted dosing instead of standard dosing

■

■ Use of sugammadex + rocuronium instead of atracurium for more reliable reversal

§ Forms inactive complex that is renally cleared, not suitable in ESRF/CKD

§ Sugammadex expensive, also forms complex with oral contraceptive to reduce efficacy

If under GA: ensure adequate depth of anaesthesia and analgesia to prevent movement, use remifentanyl infusion for immobility

§ May not be possible in surgeries requiring adequate muscle relaxation (laparoscopic) or high risk surgeries requiring absolute still surgical field (neurosurgery, endoscopic)

Increased sensitivity

§ Neuromuscular conditions (Guillain Barre syndrome, motor neurone disease, muscular dystrophy, myasthenia gravis)

§ Pharmacological: MgSO₄, calcium channel blockade

§ Electrolyte derangements: hyperMg/hypoCa/hyperK, metabolic acidosis >> correct

Use of adductor pollicis to monitor blockade

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